

Óptica II

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Capítulo 1

Quantum Ideal Gas

1.1. Non-interacting Ising model

$$H = -J \sum_i s_i \text{ where } s_i = \pm 1 \text{ (spin)}$$

We must to find:

$$Z$$

$$A$$

$$U = \langle H \rangle$$

$$\langle M \rangle = \langle \sum s_i \rangle / N$$

For interacting case

At T_c my particles are disordered. Below T_c , symmetry is broken (spontaneous magnetization).

$$Z = (z_i)^N$$

$$z_i = \sum_{s_i=\pm 1} e^{-\beta H_i} = e^{\beta J} + e^{-\beta J} = 2 \cosh(\beta J)$$

$$Z = (2 \cosh(\beta J))^N$$

$$A = -kT \ln Z = -NkT \ln(2 \cosh(\beta J))$$

$$U = \left(-\frac{\partial \ln Z}{\partial \beta} \right)_{N,V} = -\frac{\partial}{\partial \beta} (N \ln(2 \cosh(\beta J))) = -N \frac{2 \sinh(\beta J)}{2 \cosh(\beta J)} \cdot J = -NJ \tanh(\beta J)$$

In other way:

$$U = \langle H \rangle = -J \sum_i \langle s_i \rangle = -NJ \tanh(\beta J)$$
$$\langle s_i \rangle = 1 \cdot P(s_i = 1) - 1 \cdot P(s_i = -1) = \frac{2 \sinh(\beta J)}{2 \cosh(\beta J)} = \tanh(\beta J)$$

$$\text{where } P(s_i) = \frac{e^{-\beta J s_i}}{2 \cosh(\beta J)}$$
$$P(1) = \frac{e^{\beta J}}{2 \cosh(\beta J)}$$
$$P(-1) = \frac{e^{-\beta J}}{2 \cosh(\beta J)}$$

1.2. Ideal Quantum Gas

System made of N identical particles:

$$\Psi(r_1, r_2, \dots, r_N, t) \quad (1.1)$$

Its evolution is given by:

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi \quad ; \quad H = \sum_{i=1}^N \frac{p_i^2}{2m} + V(r_1, r_2, \dots, r_N) \quad (1.2)$$

P_{ij} is the **permutation operator** which exchanges the labels of particles i and j :

$$P_{ij}\Psi(\dots, r_i, \dots, r_j, \dots, t) = \Psi(\dots, r_j, \dots, r_i, \dots, t) \quad (1.3)$$

$$[H, P_{ij}] = 0 \quad \forall i, j \quad (i \neq j)$$

If $H\phi_n = E_n\phi_n$:

$$\phi(r_1, \dots, r_N, t) = \sum a_n \phi_n(r_1, \dots, r_N) \quad ; \quad a_n = \langle \phi_n | \Psi \rangle \quad (1.4)$$

Now we apply P_{ij} to Ψ and then H :

$$\begin{aligned} HP_{ij}\Psi(\dots, r_i, \dots) &\rightarrow \\ H\Psi(\dots, r_j, \dots, r_i, \dots, t) &= \sum a_n E_n \phi_n(\dots, r_j, \dots, r_i, \dots, t) = \sum a_n E_n \underbrace{P_{ij}}_{\text{scalars}} \phi_n(\dots, r_i, \dots, r_j, \dots, t) \\ &= P_{ij} \sum a_n E_n \phi_n(\dots, r_i, \dots, r_j, \dots, t) = P_{ij} H\Psi \\ &\rightarrow \text{We have proved that } [H, P_{ij}] = 0 \quad \forall i, j \quad (i \neq j) \end{aligned}$$

Commuting observables have a complete set of common eigenfunctions:

$$P_{ij}H\phi = P_{ij}E\phi = E(P_{ij}\phi) \xrightarrow{[H, P_{ij}]=0} H(P_{ij}\phi) \quad (1.5)$$

This implies $P_{ij}\phi$ is also an eigenfunction of H with the eigenvalue E .

$$P_{ij}\phi = \Theta\phi$$

Value of Θ ?

$$\begin{aligned} P_{ij}\phi(\dots, r_i, \dots, r_j, \dots, t) &= \phi(\dots, r_j, \dots, r_i, \dots, t) \\ P_{ij}^2\phi(\dots, r_i, \dots, r_j, \dots, t) &= \phi(\dots, r_i, \dots, r_j, \dots, t) = I\phi \end{aligned}$$

Then $P_{ij}^2 = I$.

As $P_{ij}\phi = \Theta\phi$:

$$\begin{aligned}P_{ij}\phi(\dots, r_i, \dots, r_j, \dots, t) &= \Theta\phi(\dots, r_i, \dots, r_j, \dots, t) \\P_{ij}^2\phi(\dots, r_i, \dots, r_j, \dots, t) &= \Theta^2\phi(\dots, r_i, \dots, r_j, \dots, t)\end{aligned}$$

As $P_{ij}^2 = I$, then $\Theta^2 = 1 \implies \Theta = \pm 1$.

- **Bosons:** $\Theta = +1 \rightarrow \text{Symmetric}$
- **Fermions:** $\Theta = -1 \rightarrow \text{Antisymmetric}$

What happens if two particles are in the same position ($r_i = r_j$) and I exchange them with P_{ij} ?

First case (Fermions): $P_{ij}\phi(\dots, r_i, \dots, r_i, \dots, t) = -\phi(\dots, r_i, \dots, r_i, \dots, t)$
 $\implies \phi(\dots, r_i, \dots, r_i, \dots, t) = 0 \rightarrow$ It is **forbidden** that two particles occupy the same position. These particles are known as **Fermions** (half-integer spin).

Second case (Bosons): $P_{ij}\phi(\dots, r_i, \dots, r_i, \dots, t) = \phi(\dots, r_i, \dots, r_i, \dots, t)$
 $\implies \phi(\dots, r_i, \dots, r_i, \dots, t) = \phi(\dots, r_i, \dots, r_i, \dots, t) \rightarrow$ It is **not forbidden** that two particles occupy the same position. These particles are known as **Bosons** (integer spin).

Examples:

Fermions

Electron

Proton

Neutron

$\vdots$

ϕ is antisymmetric ($\Theta = -1$)

Bosons

Quark

Gluon

α particle

$\vdots$

ϕ is symmetric ($\Theta = +1$)

$$\text{Ideal} \rightarrow H = \sum_i^N H_i$$

$$\begin{aligned}\sum_i n_i &= N \\ \sum_i n_i \epsilon_i &= E\end{aligned}$$

Fermions vs Bosons occupancy:

- **Fermions:** $n_j \in \{0, 1\}$ (Pauli exclusion principle).
- **Bosons:** $n_j \in \{0, 1, 2, \dots, N\}$ (No restriction).

$$Z(T, V) = \sum_{\{n_i\}} e^{-\beta \sum n_i \epsilon_i} = \sum_{\{n_i\}} e^{-\beta E(\{n_i\})} \delta_{N, \sum n_i} \quad (1.6)$$

→ This restriction ($\sum n_i = N$) does not allow us to perform the sum easily.

Boltzmann approximation → $z_N = \sum_{\{n_i\}} \frac{n_1! n_2! \dots n_i!}{N!} e^{-\beta \sum n_i \epsilon_i} \approx \frac{1}{N!} \sum_{\{n_i\}} e^{-\beta \sum n_i \epsilon_i}$

Ensembles Comparison:

- **Microcanonical:** (E, V, N) fixed. Isolated system.
- **Canonical:** ($\langle E \rangle, V, N$) fixed. Thermal contact with reservoir at T .
- **Grand Canonical (Macrocanonical):** ($\langle E \rangle, V, \langle N \rangle$) fixed. Thermal and particle contact (μ, T).

Macrocanonical ensemble

$\Xi(\mu, T, V) = \sum_{N=0}^{\infty} e^{\beta \mu N} Z_N(T, V) \rightarrow \text{similar to } \int e^{tx} p(x) dx \rightarrow \text{moment generating function}$

$$\begin{aligned} \Xi(\mu, T, V) &= \sum_{N=0}^{\infty} e^{\beta \mu N} \sum_{\{n_i\}} e^{-\beta \sum n_i \epsilon_i} \delta_{N, \sum n_i} = \sum_{\{n_i\}} e^{\beta \mu \sum n_i} \sum_{N=0}^{\infty} e^{\beta \mu N} \delta_{N, \sum n_i} \\ &= \sum_{\{n_i\}} e^{-\beta \sum n_i \epsilon_i} e^{\beta \mu \sum n_i} = \sum_{\{n_i\}} e^{\beta \sum (\mu - \epsilon_i) n_i} = \sum_{\{n_i\}} \prod_i e^{\beta (\mu - \epsilon_i) n_i} \\ &= \prod_i \sum_{\{n_i\}} e^{\beta (\mu - \epsilon_i) n_i} \end{aligned}$$

Fermions

$$n_i = 0, 1 \quad \forall i$$

then

$$\Xi(\mu, T, V) = \prod_i (1 + e^{\beta (\mu - \epsilon_i)}) \quad (1.7)$$

where 1 is when $n_i = 0$ and the exponential is when $n_i = 1$.

Bosons

$$n_i = 0, 1, \dots, \infty \quad \forall i$$

then

$$\Xi(\mu, T, V) = \prod_i \sum_{n_i=0}^{\infty} (e^{\beta(\mu-\epsilon_i)})^{n_i} = \prod_i (1 - e^{\beta(\mu-\epsilon_i)})^{-1} \quad (1.8)$$

using $\sum x^n = \frac{1}{1-x}$ for $|x| < 1$.

I can write generally this:

$$\Xi(\mu, T, V) = \prod_i (1 - \Theta e^{\beta(\mu-\epsilon_i)})^{-1/\Theta} \quad (1.9)$$

where $\Theta = \begin{cases} +1 & \text{for bosons} \\ -1 & \text{for fermions} \end{cases}$

Canonical ensemble	Macrocanonical ensemble
$A = -kT \ln Z$	$J = -kT \ln \Xi$
$A = U - TS$	$J = U - TS - \mu N$

From the macrocanonical ensemble we are going to use:

- $J = -kT \ln \Xi$
- $PV = -J$
- $\langle N \rangle = - \left(\frac{\partial J}{\partial \mu} \right)_{T,V}$

From what we know:

$$J = -kT \ln \Xi = -kT \ln \left[\prod_i (1 - \Theta e^{\beta(\mu-\epsilon_i)})^{-1/\Theta} \right] = \frac{kT}{\Theta} \sum_i \ln(1 - \Theta e^{\beta(\mu-\epsilon_i)}) \quad (1.10)$$

For Bosons $\mu < 0$ For Fermions μ can be any value

$$\langle N \rangle = - \left(\frac{\partial J}{\partial \mu} \right)_{T,V} = - \frac{kT}{\Theta} \sum_i \frac{-\Theta \beta e^{\beta(\mu-\epsilon_i)}}{1 - \Theta e^{\beta(\mu-\epsilon_i)}} = \sum_i \frac{1}{e^{\beta(\epsilon_i-\mu)} - \Theta} \rightarrow \text{this remembers us the restriction} \quad (1.11)$$

$$\Rightarrow \langle N \rangle = \langle \sum n_i \rangle = \sum \langle n_i \rangle$$

So we can know the average of n_i for each energy level ϵ_i :

$$\langle n_i \rangle = \frac{1}{e^{\beta(\epsilon_i-\mu)} - \Theta} \quad (1.12)$$

Average occupation number of level i :

- **Fermi-Dirac Statistics (Fermions):** $\langle n_i \rangle = \frac{1}{e^{\beta(\epsilon_i - \mu)} + 1}$
- **Bose-Einstein Statistics (Bosons):** $\langle n_i \rangle = \frac{1}{e^{\beta(\epsilon_i - \mu)} - 1}$

$$J = \frac{kT}{\Theta} \sum_i \ln(1 - \Theta e^{\beta(\mu - \epsilon_i)}) \quad (1.13)$$

$$\lim_{\Theta \rightarrow 0} J = \lim_{\Theta \rightarrow 0} -kT \sum_i \ln(1 - \Theta e^{\beta(\mu - \epsilon_i)})^{1/\Theta} = -kT \sum_i e^{\beta(\mu - \epsilon_i)} \quad (1.14)$$

Using $e^x = \lim_{n \rightarrow \infty} (1 + \frac{x}{n})^n$

Remember that $z_{MB} = \frac{1}{N!} (z_{MB})^N$ so $\Xi(\mu, T, V) = \sum_{N=0}^{\infty} e^{\beta\mu N} z_{MB} = \sum_{N=0}^{\infty} \lambda^N \frac{z_1^N}{N!} = e^{\lambda z_1}$ where λ is the fugacity: $\lambda = e^{\beta\mu}$

we reached the same result as the boltzmann gas approximation

$$J = -kT \ln \Xi = -\lambda z_1 kT = -kT e^{\beta\mu} \sum_i e^{-\beta\epsilon_i} = -kT \sum_i e^{\beta(\mu - \epsilon_i)}$$

$$\text{Then } J(\mu, T, V) = \frac{kT}{\Theta} \sum_i \ln(1 - \Theta e^{\beta(\mu - \epsilon_i)}) \begin{cases} \Theta = 1 & \text{Bosons} \\ \Theta = -1 & \text{Fermions} \\ \Theta \rightarrow 0 & \text{MB} \end{cases} \rightarrow \text{this uni-}$$

fied the three statistics.

$$\langle n_i \rangle = \frac{1}{e^{\beta(\epsilon_i - \mu)} - \Theta}$$

Boltzmann approximation $\rightarrow$ We see that bosons and fermions converge to the same behavior. In this limit: $\langle n_i \rangle = e^{\beta(\mu - \epsilon_i)} \ll 1$. In 3D $\rightarrow \rho \Lambda^3 \ll 1$.

Let's demonstrate that the Boltzmann approximation (which means $\langle n_i \rangle = e^{\beta(\mu - \epsilon_i)} \ll 1$) corresponds to considering $\rho \Lambda^3 \ll 1$

For free particle in 3D: $z_1 = \frac{V}{\Lambda^3}$, then $Z = \frac{z_1^N}{N!}$ (where $L \gg \Lambda$)
 $\rightarrow$ as calculated before for the Boltzmann gas.

$$A(N, T, V) = -kT \ln Z = -kT \ln \left(\frac{V^N}{\Lambda^{3N} N!} \right)$$

$$\mu = \left(\frac{\partial A}{\partial N} \right)_{T,V} \rightarrow \text{The chemical potential is the cost of Free Energy of varying } N.$$

A system always wants to minimize the Helmholtz Free Energy: $A = U - TS$

- sometimes we just need to minimize $U \rightarrow$ Example: $T = 0$
- sometimes we just need to maximize $S \rightarrow$ Example: fixed U

What does μ mean to A ?

- When $\mu < 0$, this means that as we increase N , we get a lower A . In this situation, the system wants more N in order to decrease A .
- When $\mu > 0$, this means that as we decrease N , we get higher A . Then the system wants to reduce N to decrease A .

Boltzmann approximation: $e^{\beta(\mu-\epsilon)} \ll 1 \implies \beta(\mu-\epsilon) \ll 0 \implies \mu\beta - \epsilon\beta \ll 0 \xrightarrow{\epsilon \rightarrow 0} \ln(\rho\Lambda^3) < 0 \implies \rho\Lambda^3 \ll 1$ (If it's satisfied for $\epsilon = 0$, the most unfavorable case, it's satisfied for any ϵ).

Thus, the Boltzmann approximation is equivalent to considering $\rho \ll 1$ and/or $T \gg 1$.

Thermodynamic properties of Bosons and Fermions

$$J(\mu, T, V) = \frac{kT}{\Theta} \sum_i \ln(1 - \Theta e^{\beta(\mu - \epsilon_i)})$$

We are going to consider **free particles in a 3D box**. What is the spectrum in this case?

$$\begin{aligned} H\psi_{n_x, n_y, n_z} &= \epsilon_{n_x, n_y, n_z} \psi_{n_x, n_y, n_z} \\ H &= \sum_{i=1}^N -\frac{\hbar^2}{2m} \nabla_i^2 \\ \epsilon_{n_x, n_y, n_z} &= \epsilon_{n_x} + \epsilon_{n_y} + \epsilon_{n_z} \\ \epsilon_{n_x} &= \frac{\hbar^2 \pi^2}{2m L^2} n_x^2 \quad \text{where } n = x, y, z \\ \psi_{n_x, n_y, n_z} &= \left(\frac{2}{L}\right)^{3/2} \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right) \end{aligned}$$

As $L \rightarrow \infty$, we are going to perform the sum over n_x, n_y, n_z instead of ϵ :
 $J(\mu, T, V) = \frac{kT}{\Theta} \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} \ln(1 - \Theta e^{\beta(\mu - \epsilon_{n_x, n_y, n_z})})$

$$\xrightarrow{\text{change of variable}} p_{n_x} = \frac{\hbar \pi n_x}{L} \rightarrow \epsilon_{n_x, n_y, n_z} = \frac{p_{n_x}^2 + p_{n_y}^2 + p_{n_z}^2}{2m}$$

$$\implies J(\mu, T, V) = \frac{kT}{\Theta} \sum_{\vec{n}} \ln \left(1 - \Theta e^{\beta \left(\mu - \frac{\hbar^2 \pi^2 (n_x^2 + n_y^2 + n_z^2)}{2m L^2} \right)} \right)$$